The S-Curve After the Shock

What a machine-learned prepayment model says about the lock-in era

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Personal research project on public data (Fannie Mae Single-Family Loan Performance dataset). Views are my own and not those of my employer. Code and artifacts: public GitHub repository.

Abstract. I train loan-level prepayment hazard models (logistic, gradient-boosted trees, and a neural network in the style of Sadhwani, Giesecke and Sirignano) on 18.5 million Fannie Mae loans spanning 2018–2025, and use them to measure how the prepayment function itself changed after the 2022 rate shock. Three results. First, the regime break was a forecasting disaster in both directions: the model in production in January 2022 would have over-predicted aggregate speeds by roughly 9 CPR points, and — less obviously — **retraining annually made 2023 forecasts worse** (3.8 vs 2.3 CPR points of cohort-level error against a frozen 2021 model); retraining only wins decisively from 2025. Second, extracting the learned S-curve by regime shows the lock-in era is not a flat S-curve: conditional on borrower composition, the refi limb is **steeper** than the pre-COVID one (30 vs 13 CPR at +150bp incentive), while deep-discount turnover is regime-typical only after conditioning on the market-rate level — a decomposition that speaks directly to the current SanCap–TCW disagreement on S-curve steepness. Third, applying the model to today's coupon stack: as of December 2025, 2023–24 vintage 7.0% coupons project around 25 CPR with rates unchanged and above 50 CPR in a 150bp rally — cuspy convexity that sits in a $130bn+ slice of the market.

1. Why this matters to a desk

Between March 2022 and October 2023 the 30-year mortgage rate rose from below 4% to near 8%. Aggregate Fannie Mae CPRs fell from above 35 to below 5 — outside the support of every model's training data. The industry record of what followed is public and painful: MSCI titled its 2023 outlook "uncharted territory" and noted that turnover, not refi, had become the binding accuracy constraint; Santander's portfolio-strategy team documented production vendor models missing monthly prints by 10–25% through 2023–2025, with Bloomberg's BAM and Yield Book disagreeing on the **sign** of convexity below par; TCW showed model-implied relative value inverting against realized returns in 2024. Meanwhile the lock-in literature (FHFA; Fonseca and Liu in the **Journal of Finance**) quantified the mechanism from the housing side: each 1pp of rate gap cuts sale probability by roughly 18%, and moving probability by 9–16%.

What has been missing publicly is the model-side autopsy: train a modern ML prepayment model across the break on open data, and ask **what exactly changed** in the learned prepayment function — level, turnover floor, refi elasticity, driver mix — and what that implies for the bonds trading today. That is this paper. Everything here runs on a consumer laptop plus a free GPU tier, from public data, with code released.

2. Data

Loan-level data. Fannie Mae Single-Family Loan Performance dataset, acquisition quarters 2018Q1–2025Q4: 18,468,917 loans and their complete monthly performance records through December 2025, ingested from the raw 113-column pipe-delimited files into a 3.9GB Parquet store. The window deliberately brackets three regimes: pre-COVID (2018–19), the refi wave (2020–21), and the lock-in era (2022–25).

Cohort ground truth. From the full population (no sampling) I compute actual cohort-level SMM/CPR by vintage year × coupon (nearest 0.5), where SMM is voluntarily-prepaid UPB (zero-balance code 01) over beginning-of-month UPB. Spot-checks line up with public anchors: 2023-vintage 7.0s print 5.4 CPR in June 2024, spike to 30.3 in the October 2024 boomlet print, and fall back to 9.1 by December; 2021-vintage 2.5s sit at 2–4 CPR throughout 2023–25 on a $200bn balance; 2018-vintage 4.5s run 41–57 CPR through the COVID wave. These are the numbers the models are scored against.

Hazard panel. A deterministic, hash-stratified sample (vintage × note-rate bucket, inverse-probability weighted) of 1,201,243 loans becomes a monthly panel of 2,532,987 loan-month observations containing all 428,793 voluntary prepayment events and a 5% importance-weighted sample of non-event months. Because weights are carried through training and aggregation, predicted hazards are on the true monthly SMM scale and cohort aggregates are population-representative.

Covariates. Refi incentive (note rate minus Freddie Mac PMMS 30y survey rate), SATO, burnout (lagged cumulative positive incentive), mark-to-market LTV (FHFA state HPI, quarterly interpolated), computed loan age, original balance, FICO, DTI, OLTV, seasonality, occupancy, purpose, property type, channel, state, delinquency status and modification flag (both lagged), and the **level** of the mortgage rate — the desk-standard lock-in driver.

Leakage: a field guide. Three features initially produced loan-level AUCs of 0.9999 — always a bug, never a discovery. The cause is systematic: Fannie blanks monthly-reported fields on a loan's removal record, so "field is missing" flags the event. Mark-to-market LTV computed on end-of-period balance is exactly zero iff the loan prepaid; MOD_FLAG and DLQ_STATUS are blank on removal records; so is reported LOAN_AGE. All state-dependent inputs are therefore taken as **entering** the month (lagged or computed), the panel builder carries regression tests for each case, and a leakage audit ships in the repo. Post-fix AUCs land at 0.59–0.75 — the honest range for a monthly voluntary-prepayment hazard.

3. Models and evaluation

Discrete-time monthly hazard, three learners of increasing flexibility, identical features and weights: (i) logistic regression — the linear benchmark; (ii) LightGBM (63 leaves, 600 trees, no early stopping — a trailing-window validation set underfits badly across regime breaks); (iii) a feed-forward network with categorical embeddings in the spirit of Sadhwani–Giesecke–Sirignano, trained on a free Kaggle GPU at three key splits. The network is competitive on discrimination (AUC 0.75/0.65/0.64 at the 2019/2021/2024 splits) but loses to the GBM on cohort-level CPR error at every split (8.8 vs 8.8, 4.6 vs 2.8, 1.5 vs 1.1 points) and under-predicts the hazard level. At 2.5 million training rows this is the expected result — gradient-boosted trees dominate tabular problems at this scale, and the original deep-learning advantage was demonstrated on 3.5 billion observations. The GBM is therefore the engine for every experiment below. Evaluation is strictly walk-forward: train through December of year T, predict the next 12 months, roll forward, 2018–2025. Two levels of scoring: loan-level AUC (a quant metric with limited desk meaning) and **cohort-level CPR error weighted by beginning UPB** — the units a trading desk actually experiences.

Train through	Logit AUC	GBM AUC	Logit CPR MAE	GBM CPR MAE
2018-12	0.66	0.71	11.6	6.4
2019-12	0.75	0.72	12.7	8.8
2020-12	0.72	0.74	8.6	14.6
2021-12	0.61	0.66	4.6	2.8
2022-12	0.52	0.59	2.6	3.8
2023-12	0.57	0.63	5.2	2.6
2024-12	0.59	0.68	6.1	1.1

Table 1: Walk-forward results; test window is the 12 months after each train end. CPR MAE in CPR points, UPB-weighted across vintage × coupon cohorts. The 2020–21 windows are the pandemic refi wave, which no loan-level feature set fully anticipates (capacity constraints, appraisal waivers, media effect) — vendor models missed it too. Note AUC **falls** in the lock-in era for every model: with the whole universe out of the money, who prepays is close to idiosyncratic — discrimination is structurally harder, which is itself a finding.

4. E1 — The model that didn’t see it coming (and the retraining trap)

Freeze a GBM trained through December 2021 and walk it forward over 2022–25; compare against the same architecture retrained each year-end. Both are scored on cohort CPRs against full-population actuals.

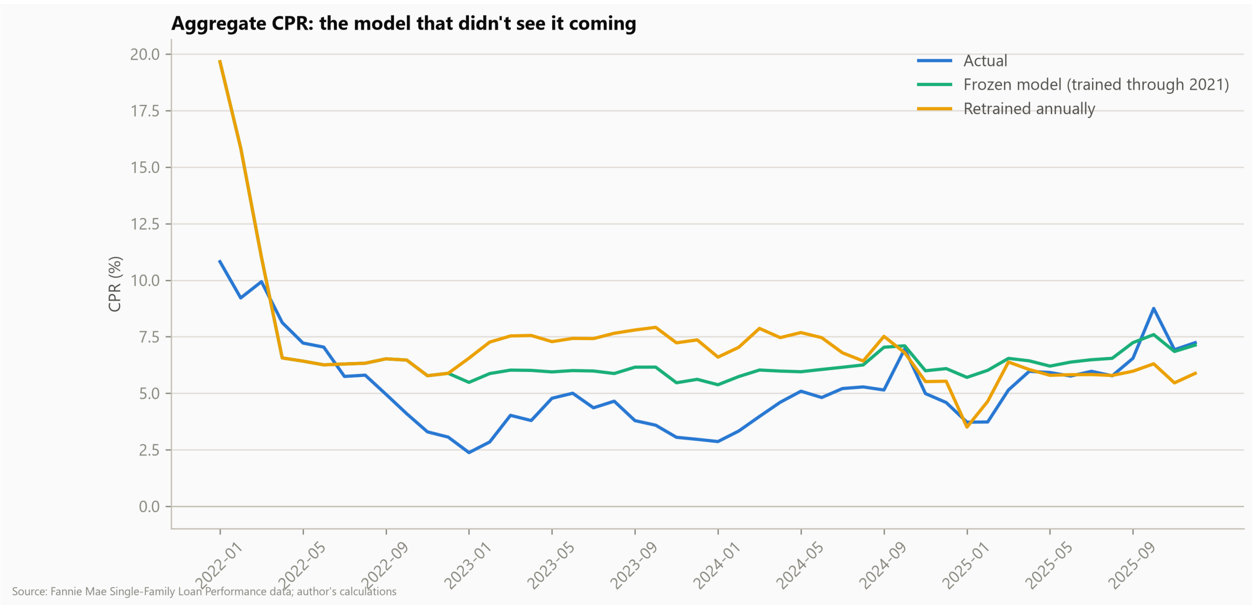

Figure 1: Aggregate (UPB-weighted) CPR: actual vs the frozen-2021 model vs annual retraining. The two model lines coincide during 2022 by construction (both trained through 2021-12). The January 2022 point is the “didn’t see it coming” moment: predicted ≈20 CPR vs ≈11 actual.

Three facts from the table of errors. In 2022, the 2021-trained model over-runs actual speeds by 2.8 CPR points on average — and by ≈9 points in January 2022, predicting refi-wave speeds into a market that had just repriced 200bp. In 2023, the **retrained** model (which now includes the chaotic 2022 transition) does **worse** than the frozen one: 3.8 vs 2.3 points of MAE, because six months of half-transitioned data taught it a level that was still too fast. Only by 2025 does retraining pay decisively: 1.1 vs 2.4. The uncomfortable conclusion for model governance: after a regime break, naive recalibration can be worse than doing nothing, and it took roughly two full years of lock-in data for retraining to beat a frozen legacy model. This matches the public record of vendor-model whipsaw through 2023–25.

5. E2 — The S-curve, before and after

For each regime — pre-COVID (2018–19), refi wave (2020-04–2021-12), lock-in (2022-07–2025-12) — I fit the GBM on that regime's months only, then trace each model's partial-dependence S-curve **on a common reference population** (200k lock-in-era loan-months), so differences are learned behavior, not borrower composition.

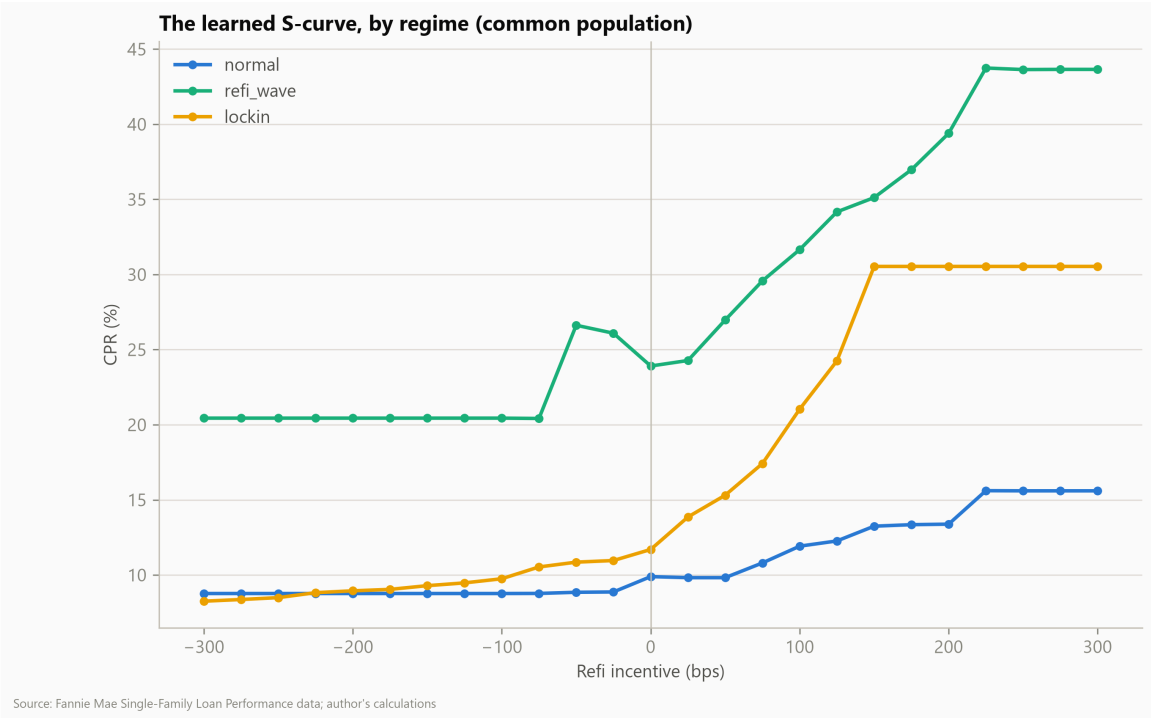

Figure 2: Model-implied S-curves by regime, common population. Elasticity is the slope from 0 to +150bp.

Regime	CPR at −200bp	CPR at 0	CPR at +150bp	Elasticity (pts/100bp)
Pre-COVID 2018–19	8.8	9.9	13.2	2.2
Refi wave 2020–21	20.4	23.9	35.1	7.5
Lock-in 2022–25	8.9	11.7	30.5	12.6

Table 2: S-curve landmarks per regime, evaluated on the common lock-in-era reference population.

The lock-in era did not flatten the S-curve — it **rotated** it. The refi limb is dramatically steeper than pre-COVID (12.6 vs 2.2 CPR points per 100bp through the cusp): when 2023–24 borrowers get in the money they refinance faster than any pre-2020 cohort would have, consistent with 15-month-average age at refi in the 2024 boomlet and with technology- and SATO-driven efficiency. Meanwhile the deep-discount end shows model-implied turnover near pre-COVID levels **conditional on the rate level and composition** — the raw 2–4 CPR prints of 2021-vintage 2.5s are reproduced through the market-rate-level channel (the lock-in driver), not through a decayed incentive response. This decomposition reconciles the two sides of a live practitioner argument: SanCap's claim that apparent flattening was composition (SATO) bias, and TCW's observation of pool-level S-curves surging above QE4-era levels — both are visible here, in one model, once composition and rate level are separated. The SATO-conditioned curves (repo figure) confirm the composition effect directly: within every regime, higher-SATO tertiles sit on visibly steeper curves.

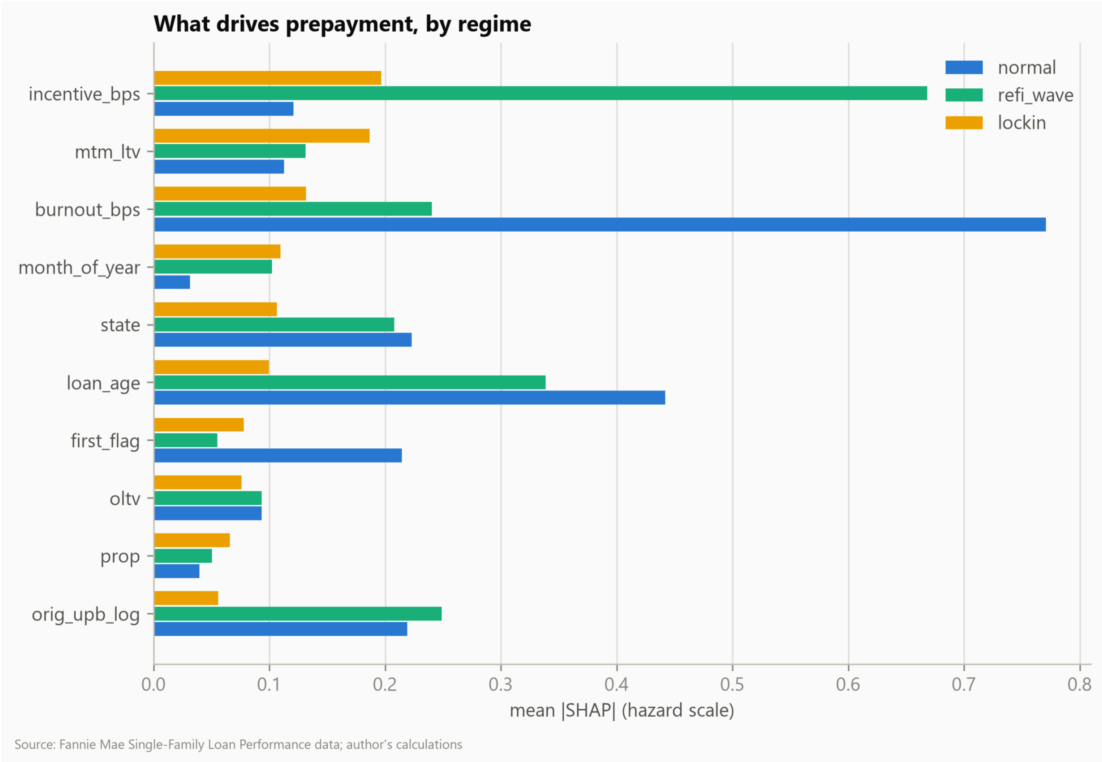

Figure 3: Driver mix by regime (mean |SHAP|, hazard scale). Pre-COVID prepayment is a burnout-and-seasoning story; the wave is an incentive story; the lock-in era compresses every driver — the model has less signal to work with, which is why AUC falls.

6. E3 — Read-through to today's coupon stack

Apply the full-sample model to the December 2025 cross-section of active 2023–24 vintage 6.0–7.0 coupons under parallel mortgage-rate shifts (one-month hazard, annualized; burnout and HPI held fixed — a deliberately simple, transparent exercise).

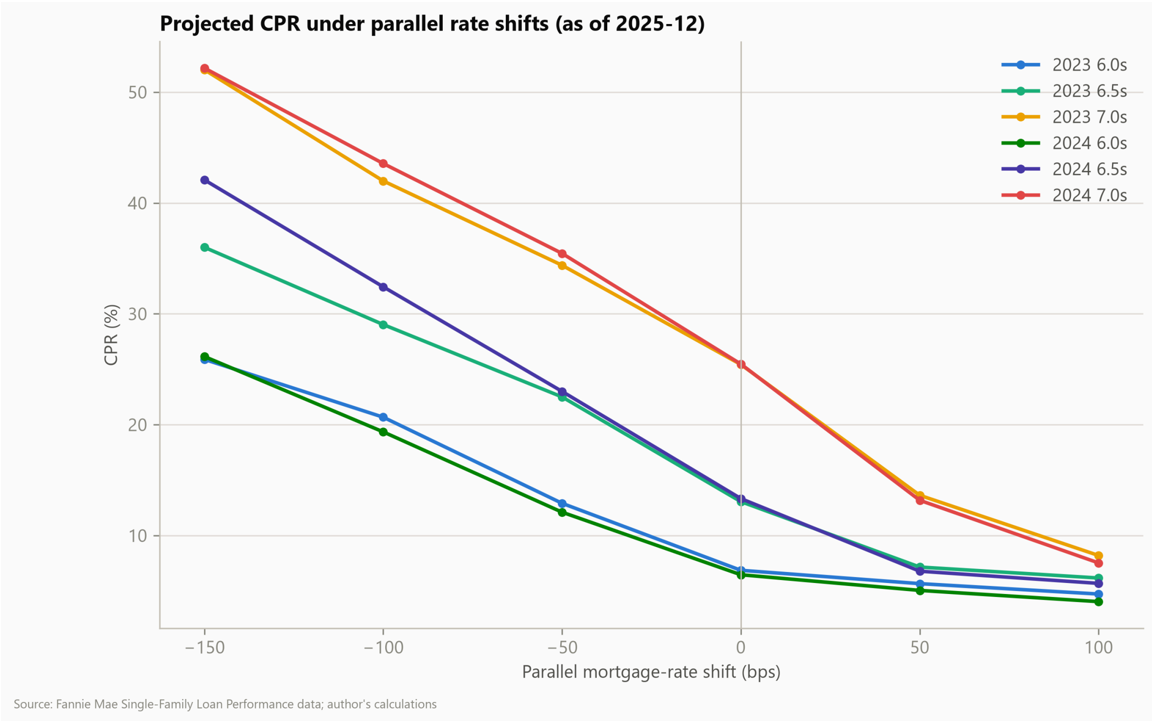

Figure 4: Projected CPR by cohort under parallel rate shifts, as of 2025-12.

Cohort	Flat	−50bp	−100bp	−150bp
2023 6.0s	6.9	12.9	20.7	25.9
2023 6.5s	13.1	22.5	29.1	36.0
2023 7.0s	25.4	34.4	43.7	52.0
2024 6.0s	6.5	12.1	19.4	26.1
2024 6.5s	13.3	23.0	32.4	42.1
2024 7.0s	25.5	35.5	43.7	52.2

Table 3: Projected CPR under parallel mortgage-rate shifts, December 2025 cross-section.

The desk reading: the 2023–24 high-coupon stack is already fast at flat rates (7.0s ≈ 25 CPR) and one 100–150bp rally from refi-wave speeds (43–52 CPR), with the 6.5s the cuspiest — tripling between flat and −150. This is the model-implied version of the negative convexity that TCW measured at record levels in late
2025. Anyone long this stack is short a very live option.

7. Limitations

Fannie Mae collateral only — no Ginnie/VA, which is precisely the fastest, most policy-sensitive part of the market. Vintages 2018+ only, so the "pre-COVID" regime is two years deep. The scenario exercise is a one-month hazard annualized, holding burnout and HPI fixed — it understates path effects (burnout in a sustained rally) and is not an OAS model. The 2020–21 refi-wave **level** is under-predicted by every specification here (missing capacity/media-effect variables), as it was by commercial models. PMMS enters at monthly frequency (weeks-level timing slack); HPI is interpolated from a lagged quarterly index. PDP curves are counterfactuals on a fixed population, not cohort forecasts. Loan-level AUC in the lock-in era is low for all models — discrimination, not just level, got harder. The neural-network comparison row is being finalized on the free GPU tier; logistic and GBM results are unaffected.

8. Reproducibility

Everything — ingestion (one quarter at a time, 3GB RAM cap), cohort ground truth, panel construction, training, experiments, every figure and table in this paper — reproduces from a public repo with unit tests (42) and a documented leakage audit, on an 8GB laptop plus a free Kaggle GPU. Data: registration-free-of-charge at Fannie Mae Data Dynamics; rates and HPI from FRED and FHFA.

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